SRM TRP ENGINEERING COLLEGE · DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

MINI PROJECT REPORT

Bank Customer Churn Analysis

Exploratory Data Analysis, Visualization and Data Preprocessing

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OVERVIEW

Agenda

From raw customer records to a set of cautious, data-backed observations.

01 Introduction and objectives

02 Dataset and methodology

03 Data preprocessing

04 Exploratory analysis and visualizations

05 Key findings and recommendations

06 Limitations, future work and conclusion

INTRODUCTION

Why customer churn matters

BACKGROUND

Churn occurs when a customer leaves the bank. A high churn rate reduces revenue and raises the cost of acquiring replacements, because retaining a customer is generally cheaper than winning a new one.

PROBLEM STATEMENT

Raw customer records must be checked and summarized before they can support retention decisions: inspect data quality, measure the churn rate, and see how churn varies by geography, gender and age.

SCOPE Data preprocessing and exploratory analysis of 10,000 records. No trained prediction model is built.

INTRODUCTION

Project objectives

1. Inspect the dataset: shape, data types, missing values and duplicate records.
2. Measure the overall churn rate and the balance between retained and churned customers.
3. Compare churn across geography (France, Spain, Germany) and gender.
4. Compare the age distribution of retained and churned customers using a box plot.
5. Use a correlation heatmap to examine relationships among numeric features and with churn, and document the limitations.

DATASET

The SmartBank customer dataset

10,000
customer records

12
columns, 1 identifier, 10 features, 1 target

0
missing values in every column

0
duplicate rows

TARGET VARIABLE

Exited: 0 = Retained, 1 = Churned

SOURCE

SmartBank – Integrated Banking Management and Customer Churn System project

Attribute description

Attribute	Description	Role
CustomerID	Unique customer identifier	Identifier
CreditScore	Credit score of the customer	Numeric feature
Geography	Country of the customer (France, Spain, Germany)	Categorical feature
Gender	Gender of the customer (Male, Female)	Categorical feature
Age	Customer age in years	Numeric feature
Tenure	Years the customer has been with the bank	Numeric feature
Balance	Account balance of the customer	Numeric feature
NumOfProducts	Number of bank products held	Numeric feature
HasCrCard	Has a credit card (1 = Yes, 0 = No)	Binary feature
IsActiveMember	Is an active member (1 = Yes, 0 = No)	Binary feature
EstimatedSalary	Estimated salary of the customer	Numeric feature
Exited	Churn status (0 = Retained, 1 = Churned)	Target label

METHODOLOGY

A nine-step project workflow

1. Load the dataset into a Pandas DataFrame in Colab
2. Inspect shape, columns, data types and samples
3. Check for missing values and duplicate rows
4. Store the validated data as clean_df
5. Plot the overall churn distribution
6. Compare churn by geography and gender
7. Compare age of retained and churned customers
8. Generate a correlation heatmap with the churn label
9. Interpret cautiously and record limitations

TOOLS

Python in Google Colab | Pandas | NumPy | Matplotlib | Seaborn | GitHub

DATA PREPROCESSING

The data arrived clean

Inspection

Shape, column names, data types and the first five records were reviewed: 10,000 rows and 12 columns.

Missing values and duplicates

A column-wise count of missing values and a count of duplicate rows both returned zero.

Cleaning

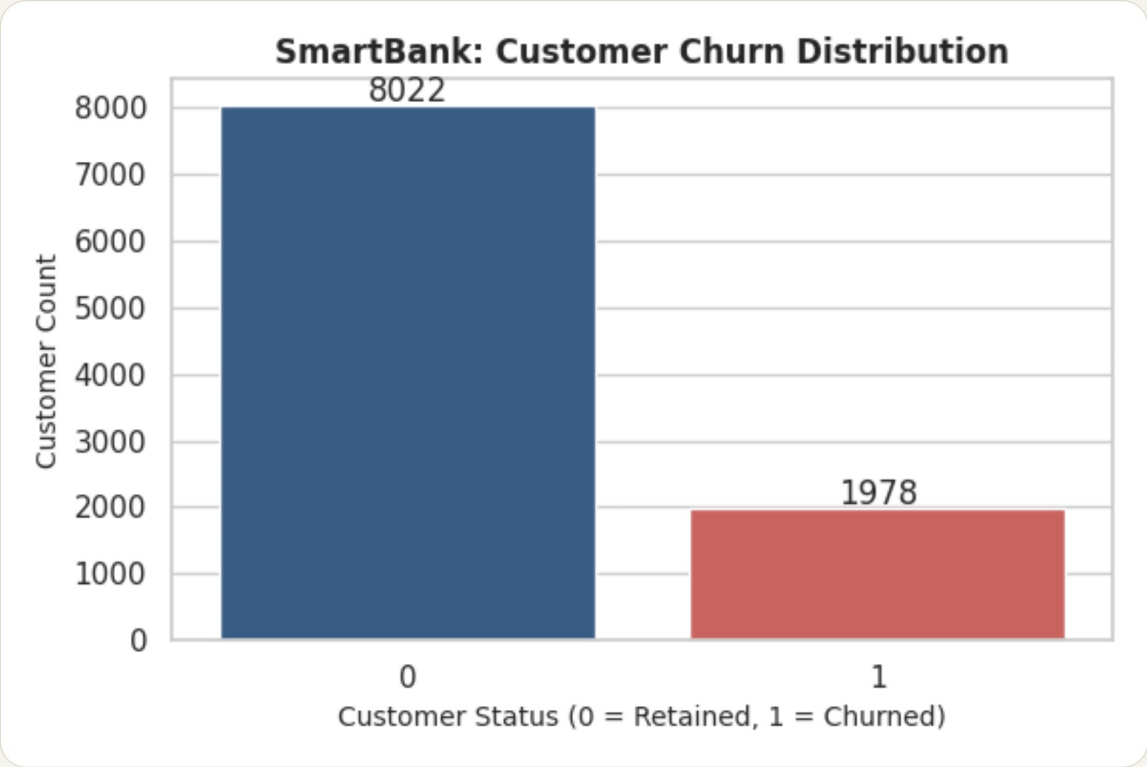
No rows were removed and no imputation was needed. clean_df keeps all 10,000 records and 12 columns.

Variable handling

Geography and Gender are plotted as categories. The 0/1 flags enter the heatmap as numbers. CustomerID is not used as a feature.

VISUALIZATION 1 · CHURN DISTRIBUTION

Roughly one in five customers churned



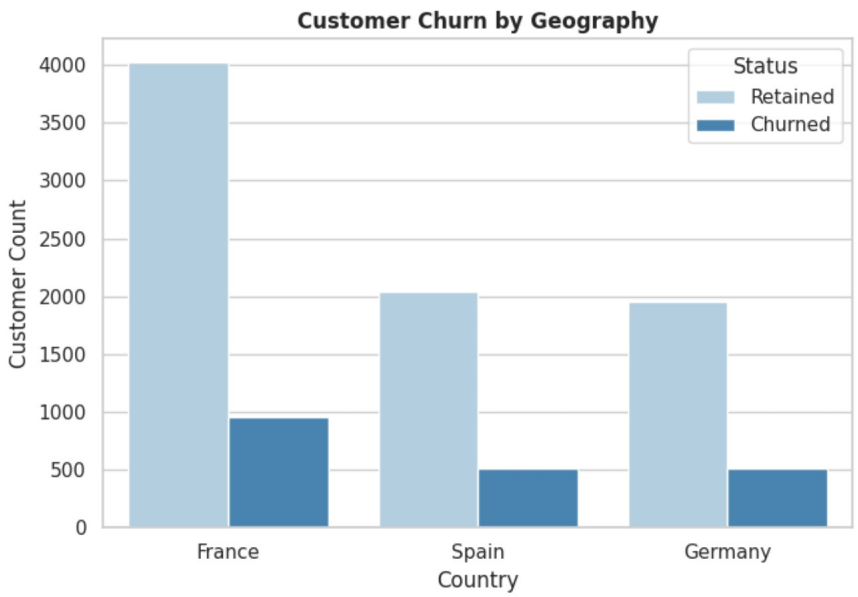
8,022
retained customers (80.22%)

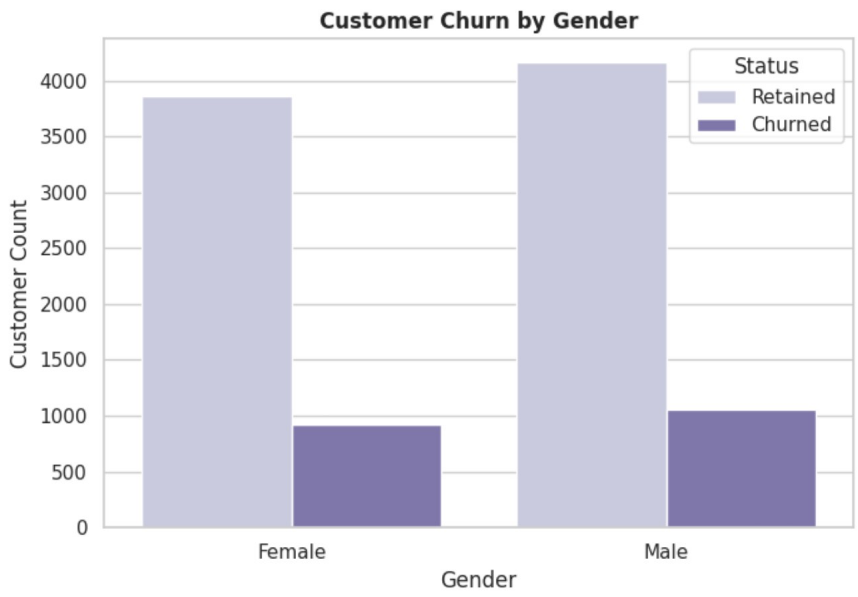
1,978
churned customers (19.78%)

The classes are imbalanced, about 4.06 : 1. Later models should be judged on precision, recall and F1-score rather than accuracy alone.

VISUALIZATION 2 · GEOGRAPHY AND GENDER

Churn is similar across groups



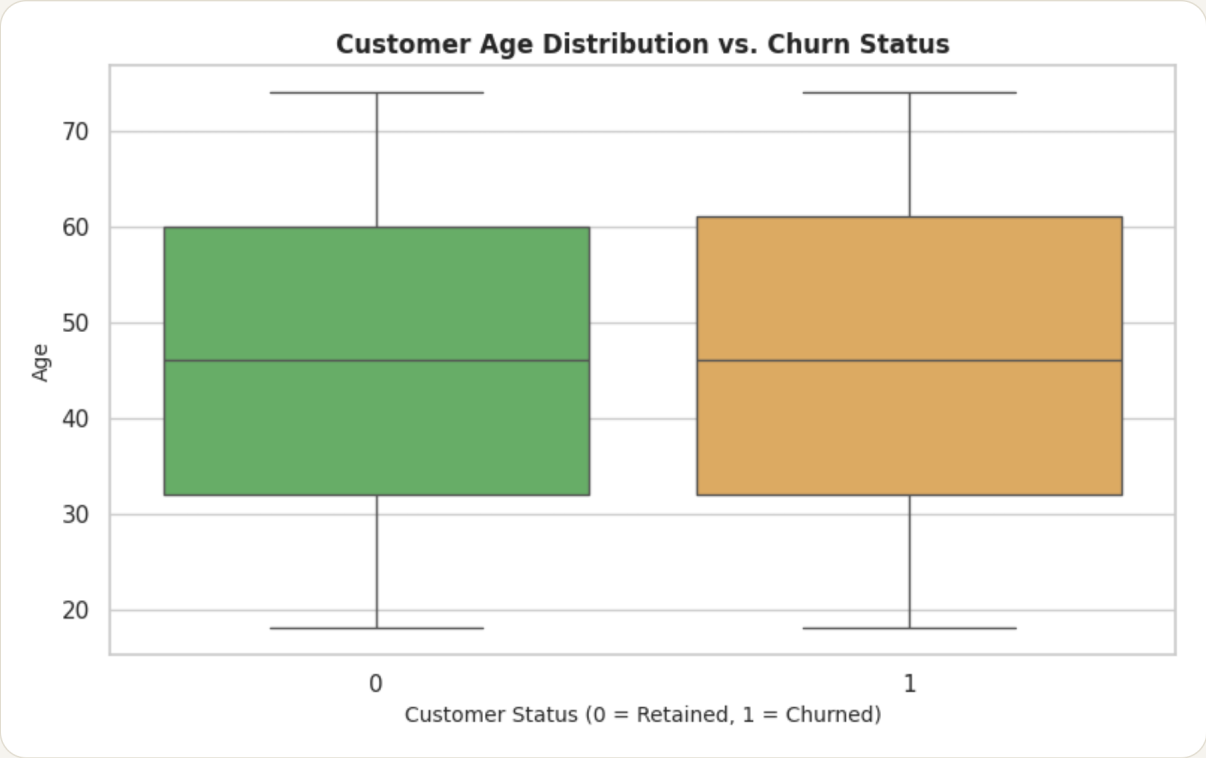


Geography. France holds about half of all customers. Churn is close to 20% in every country.

Gender. Slightly more male customers, yet churn is close to 20% in both groups.

VISUALIZATION 3 · AGE VERSUS CHURN

Age does not separate the two groups



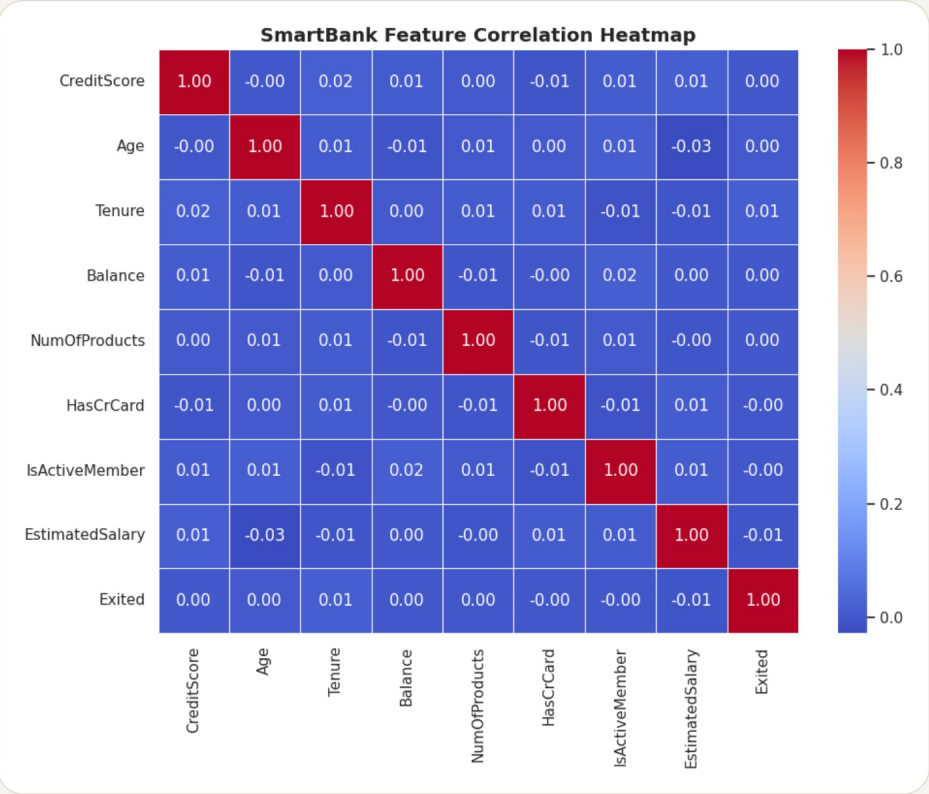
46
median age in both groups, about

32–60
interquartile range, about

18–74
overall age range in both groups

VISUALIZATION 4 · CORRELATION HEATMAP

No feature is linearly tied to churn



±0.03

largest size of any off-diagonal coefficient

0.00–0.01

size of Exited's correlation with every feature

Strongest pairs. Age with EstimatedSalary (–0.03), CreditScore with Tenure (0.02) and Balance with IsActiveMember (0.02).

Read with care. Correlation measures linear association, not causation. A value near zero does not rule out non-linear or interaction effects.

KEY FINDINGS

What the data shows

10,000	customer records and 12 variables, with no missing values and no duplicates.
19.78%	churned (1,978 customers) and 80.22% were retained, a moderately imbalanced target.
≈ 20%	churn in every country and for both genders in the displayed charts.
≈ 46 yrs	median age for both retained and churned customers, with nearly identical distributions.
±0.03	is the largest pairwise correlation in size, including every correlation with churn.

RECOMMENDATIONS

Where to go next with the data

Non-linear models	Linear correlations with churn are negligible, so test decision trees, random forests or gradient boosting, and feature interactions.
Class imbalance	Use stratified splitting, class weights or resampling, and judge models by precision, recall, F1-score, ROC-AUC and a confusion matrix.
Data dictionary	Maintain a dictionary describing every field, its unit and its allowed values.
Behavioral data	Add transaction frequency, complaints and service usage, which are more likely to explain churn than static profile fields.
Validation	Confirm findings on real, representative customer data and obtain business review before acting on any result.

LIMITATIONS AND FUTURE WORK

What this analysis cannot tell us

Limitations

- Exploratory and descriptive only: no trained or evaluated model.
- Pearson correlation captures linear links only; Geography and Gender are not in the heatmap.
- A single snapshot with no transaction history or time dimension.
- Near-zero correlations suggest a simulated or benchmark dataset, so patterns may not generalize to a real bank.
- Group differences were read visually; no significance tests were run.

Future enhancements

- Train and compare logistic regression, random forest and gradient boosting with a train/test split and cross-validation.
- Engineer features such as balance-to-salary ratio, age groups and product-usage indicators; encode categorical variables.
- Apply chi-square and t-tests to check whether group differences are significant.
- Explain predictions with feature importance and build a retention dashboard for bank managers.

CONCLUSION

A sound exploratory foundation, not yet a prediction system

The SmartBank data is clean and complete. About 19.78% of customers churned, churn was similar across countries, genders and ages, and no numeric feature showed a meaningful linear correlation with churn. These findings are exploratory only.

A dependable churn model needs non-linear modeling, richer behavioral data, proper handling of class imbalance and validation on representative customer data.

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Thank you

Questions and discussion

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Notebook: github.com/814725243064-jxc/SmartBank---Integrated-Banking-Management-Customer-Churn-System

Documentation: pandas.pydata.org · numpy.org · matplotlib.org · seaborn.pydata.org